

DAY 2 SUMMER TRAINING ASSIGNMENT

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Q1. What is a conditional statement in programming?

A conditional statement is a programming statement that allows a program to make decisions based on a condition. It checks whether a condition is true or false and executes the appropriate block of code. This helps programs respond differently for different situations.

Q2. What does an if statement do?

An if statement checks a given condition before executing code. If the condition is true, the statements inside the if block are executed. If the condition is false, the block is skipped.

Q3. What is the purpose of an else block?

An else block provides an alternative set of statements when the if condition is false. It ensures that one of the two blocks is executed. This makes decision making more complete.

Q4. When is an else block executed?

The else block runs only when the condition inside the if statement evaluates to false. It is used to handle all remaining cases that do not satisfy the if condition.

Q5. What is a Boolean expression?

A Boolean expression is an expression that produces either true or false as its result. It is commonly used in conditions with comparison or logical operators.

Q6. Write the syntax of a simple if statement.

Syntax:

```
if(condition){  
    // statements  
}
```

Q7. What operator is commonly used to test equality?

The equality operator is `==`. It compares two values and returns true if both values are equal, otherwise it returns false.

Q8. What does the modulus (%) operator return?

The modulus operator (%) returns the remainder after division of one number by another. It is commonly used to check even and odd numbers.

Q9. How can you check if a number is even?

A number is even if `number % 2 == 0`. This means the remainder after dividing by 2 is zero.

Q10. How can you check if a number is odd?

A number is odd if $\text{number} \% 2 \neq 0$. In this case the remainder after dividing by 2 is not zero.

Q11. What is the purpose of else-if?

else-if is used to check multiple conditions one after another. When one condition becomes true, its block executes and the remaining conditions are skipped.

Q12. What is a nested if statement?

A nested if statement is an if statement written inside another if statement. It is useful when one condition depends on another condition.

Q13. Can an if statement exist without an else?

Yes. An if statement can be used without an else block. If the condition is false, the program simply moves to the next statement.

Q14. What is the output when a false condition is tested in an if statement without else?

When there is no else block, nothing is printed or executed for a false condition. The program continues with the next instruction.

Q15. What is the role of braces `{ }` in C/C++ conditionals?

Braces define the beginning and end of a block of code. They improve readability and allow multiple statements to be executed together.

Q16. What does `>` mean?

`>` means greater than.

Q17. What does `<` mean?

`<` means less than.

Q18. What does `>=` mean?

`>=` means greater than or equal to.

Q19. What does `<=` mean?

`<=` means less than or equal to.

Q20. What does `!=` mean?

`!=` means not equal to.

Q21. What is a switch statement?

A switch statement is a selection statement used to choose one block of code from multiple constant choices. It makes code cleaner than many if-else statements in such cases.

Q22. What is a case label?

A case label represents a specific value inside a switch statement. If the switch expression matches that value, the corresponding statements are executed.

Q23. Why is break used in switch?

break stops the execution of the switch statement after a matching case. It prevents the program from executing the next cases.

Q24. What happens if break is omitted?

If break is omitted, execution continues into the next case. This behavior is called fall-through.

Q25. What is the default case?

The default case executes when none of the case labels match the given value. It works like the final else block.

Q26. Can switch compare strings in C++?

No. Traditional C++ switch statements cannot compare strings directly. They work mainly with integer, character, and enumeration values.

Q27. Which is better for many constant choices?

Switch is generally better when there are many constant values to compare. It improves readability and execution speed in many situations.

Q28. How do you test whether a value is positive?

Use the condition `value > 0`.

Q29. How do you test whether a value is negative?

Use the condition `value < 0`.

Q30. How do you test whether a value is zero?

Use the condition `value == 0`.

Q31. Write a condition to check if age is at least 18.

`age >= 18`

Q32. Write a condition to check if marks are above 50.

`marks > 50`

Q33. What is decision making in programming?

Decision making is the process of choosing different actions depending on conditions. It allows programs to solve real-world problems logically.

Q34. Can nested if statements have multiple levels?

Yes. Nested if statements can have multiple levels. However, too many levels can reduce readability and should be avoided when possible.

Q35. What is logical AND (&&)?

Logical AND (&&) returns true only if both conditions are true.

Q36. What is logical OR (||)?

Logical OR (||) returns true if at least one condition is true.

Q37. What is logical NOT (!)?

Logical NOT (!) reverses the result of a condition. It changes true to false and false to true.

Q38. How do you check if a year is divisible by 4?

Use `year % 4 == 0`.

Q39. What is a leap year?

A leap year contains 366 days instead of 365 days. It usually occurs every four years with some additional century rules.

Q40. What is the first condition for a leap year?

The first condition is that the year should be divisible by 4.

Q41. How do you find the largest of two numbers?

Compare the two numbers using an if statement. If `a > b` then a is larger; otherwise b is larger.

Q42. How do you find the largest of three numbers?

Use multiple if-else conditions or nested if statements to compare all three values and print the largest one.

Q43. What is an if-else-if ladder?

An if-else-if ladder is a series of conditions checked from top to bottom. The first true condition executes its block.

Q44. Difference between if and switch?

if can test complex conditions and ranges, while switch compares only constant values. Switch is usually shorter for menu-based programs.

Q45. Can switch use ranges directly?

No. Switch cannot directly check ranges like 1-10. Separate cases or if-else statements are required.

Q46. What is fall-through in switch?

Fall-through means execution continues into the next case because break was not used.

Q47. What data type is usually used for conditions?

Conditions usually evaluate to the bool data type. Comparison operators return either true or false.

Q48. Why are conditional statements important?

Conditional statements allow programs to make decisions and perform different actions depending on user input or data. They make software intelligent and interactive.

Q49. Give one real-life example.

Example: If a traffic signal is green, vehicles move; otherwise they stop. This is similar to decision making in programming.

Q50. Name four conditional structures covered.

The four conditional structures are: if, if-else, else-if ladder, nested if, and switch statement.